30-07-2024

(CO2)

[Analytical]

Assessment Test -1

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[Operating Systems]

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Questions

① CPU Scheduling

Process	Burst Time (ms)
P_1	10
P_2	4
P_3	6
P_4	2

Arrival Time =0

ⓐ SJF

ⓑ Avg. WT & Avg. TAT

ⓒ SJF to FCFS

② Round Robin Scheduling

Process	Burst Time
P_1	20
P_2	5
P_3	13
P_4	8

time Quantum = 4ms

ⓐ Gantt chart

ⓑ WT & TAT

ⓒ changing Quantum to 10ms. How would affect system Performance & Context switching.

③ Deadlock - Banker's Algorithm

3 Resource Types

A = 10 ; B = 5 ; C = 7   Total = (10, 5, 7)

The Allocation & Maximum Need matrices are

	A	B	C	Maximum matrix		
P_1	0	1	0	7	5	3
P_2	2	0	0	3	2	2
P_3	3	0	2	9	0	2
P_4	2	1	1	4	2	2
P_5	0	0	2	5	3	3

Avail = Total - Alloc

Need = max - Alloc

Need ≤ avail if true

new avail = avail + alloc

Solutions

Process	BT	AT	CT
P_1	10	0	22
P_2	4	0	6
P_3	6	0	12
P_4	2	0	2

① Shortest Job First (SJF)

P_4	P_2	P_3	P_1
0 – 2	2 – 6	6 – 12	12 – 22

ⓐ Turn Around Time = Completion Time - Arrival Time

Waiting Time = TAT - BT

TAT	WT
P_1 = 22	12
P_2 = 6	2
P_3 = 12	6
P_4 = 2	0

[∴ For each Process]

(b) Avg. TAT = $\frac{22 + 6 + 12 + 2}{4}$ Avg. WT = $\frac{12 + 2 + \text{[illegible]}}{4}$

$= \frac{\cancel{42}^{10.5}}{\cancel{4}}$ $= \frac{20}{\cancel{4}}$

$\boxed{\text{Avg. WT} = 5\text{ms}}$

$\boxed{\text{Avg. TAT} = 10.5\text{ms}}$

(c) First Come First Serve

CT
10
14
20
22

Gantt chart

P1	P2	P3	P4

0 10 14 20 22

~~FA~~ Process	BT	TAT = CT − AT	WT = TAT − BT
P1	10	10	0
P2	4	14	10
P3	6	20	14
P4	2	22	20

10

Avg. TAT = $\frac{10 + 14 + 20 + 22}{4}$

$= \frac{\cancel{66}^{16.5}}{\cancel{4}}$

$\boxed{\text{Avg. TAT} = 16.5\text{ms}}$

Avg. WT = $\frac{0 + 10 + 14 + 20}{4}$

$= \frac{44}{4}$

$\boxed{\text{Avg. WT} = 11\text{ms}}$

∴ Comparing to SJF & FCFS,

* Shortest Job First takes short time for scheduling a Process. TAT of SJF is low.

* Choosing SJF is better Option.

③ Deadlock

A=10,

Available = Total - Alloc

* Available matrix $\begin{pmatrix} A & B & C \\ 10-5, & 5-2, & 7-5 \end{pmatrix}$

Process	A	B	C
P_1	10	4	7
P_2	8	5	7
P_3	7	5	5
P_4	8	4	6
P_5	10	5	5

(5,3,2)

ⓐ Need matrix = max - Alloc

Process	A	B	C
P_1	7	4	3
P_2	1	2	2
P_3	6	0	0
P_4	2	1	1
P_5	5	3	1

ⓑ ~~Need~~ Check safe or Not

ⓑ at P_1 Check Need ≤ available if True

(7,4,3) ≤ (5,3,2) ✗

new avail = avail + alloc

P_1 (7, 4, 3) ≤ (10, 4, 7) ✓

new avail = (10, 5, 7)

P_2 (1, 2, 2) ≤ (10, 5, 7) ✓

new avail = (12, 5, 7)

P_3 (6, 0, 0) ≤ (12, 5, 7) ✓

new avail = (15, 5, 9)

P_4 (2, 1, 1) ≤ (15, 5, 9) ✓

new available = (17, 6, 10)

P_5 (5, 3, 1) ≤ (17, 6, 10) ✓

new available = (17, 6, 12)

Safest route is

$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5$

ⓒ Both Given and safest is same

P_1, P_2, P_3, P_4, P_5.

Round - Robin Scheduling

Process	BT
P_1	20
P_2	5
P_3	13
P_4	8

[time Quantum = 4ms]

a) Gantt Chartt

P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_4	P_1	P_3	P_1	P_3	P_1

0 4 8 12 16 20 21 25 29 33 37 41 42 40

(b)

Process	BT	TAT	WT
P_1	20	46	26
P_2	5	21	16
P_3	13	42	29
P_4	8	29	21

10

$$\text{Avg.TAT} = \frac{46+42+21+29}{4} = 23$$

(c) Quantum = 10ms

P_1	P_2	P_3	P_4	P_1	P_3

0 10 15 23 33 43 46

[∴ changing to Quantum 10ms improves CPU utilization.]

Process	BT	TAT	WT
P_1	20	43	23
P_2	5	15	10
P_3	13	46	33
P_4	8	33	25

$$\text{Avg.WT} = \frac{23+10+33+25}{4} = 22.75$$

$$\text{Avg.TAT} = \frac{43+15+46+33}{4} = 34.25$$

Question 4: Deadlock Detection – Resource Allocation Graph (RAG)

Given,

- P_1 is holding R_1 and waiting for R_2
- P_2 is holding R_2 and waiting for R_3
- P_3 is holding R_3 and waiting for R_1

ⓐ **Resource Allocation Graph (text description):**

In RAG,

Process $\longrightarrow$ Resource means request edge (waiting)

Resource $\longrightarrow$ Process means allocation edge (holding)

Nodes:

Processes : P_1 , P_2 , P_3

Resources : R_1 , R_2 , R_3 (each single instance)

Edges:

$R_1 \longrightarrow P_1$ (P_1 is holding R_1)

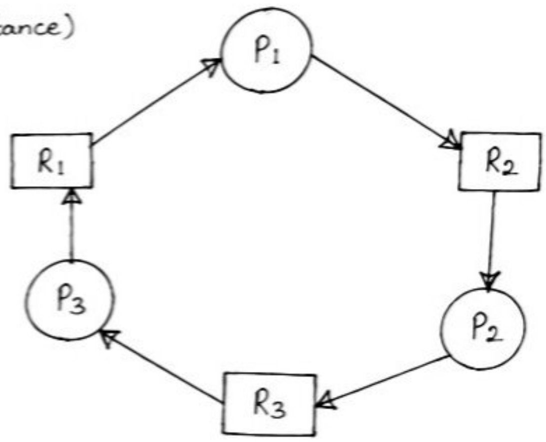
$P_1 \longrightarrow R_2$ (P_1 is waiting for R_2)

$R_2 \longrightarrow P_2$ (P_2 is holding R_2)

$P_2 \longrightarrow R_3$ (P_2 is waiting for R_3)

$R_3 \longrightarrow P_3$ (P_3 is holding R_3)

$P_3 \longrightarrow R_1$ (P_3 is waiting for R_1)



ⓑ **Is the system in deadlock?**

Yes. The RAG contains a cycle:

$$P_1 \rightarrow R_2 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1 \rightarrow P_1$$

Since each resource has only one instance and all are involved in the cycle, the system is in deadlock.

ⓒ **Recommended Strategy and Justification:**

Strategy: Deadlock Detection and Recovery

Justification:

* The system can allow resource allocation freely and detect deadlock when it occurs using cycle detection in RAG.
* After detection, recover by terminating one process (e.g., P_3), then reclaim its resources (release R_3) and proceed.
* This avoids restricting resource utilization too much (better than prevention).

Question 5: CPU Utilization – Multiprogramming Level Analysis

Given,

- Each process spends 20% time in CPU and 80% waiting for I/O.
- Degree of multiprogramming = 4 processes

CPU Utilization formula:

$$U = 1 - (p^n)$$

where,

p = probability a process is waiting for I/O

n = number of processes in memory

Here,

$$p = 80\% = 0.8$$

ⓐ **Calculate CPU Utilization for $n = 4$.**

$$U = 1 - (0.8)^4$$
$$= 1 - 0.4096$$
$$= 0.5904$$

$$\boxed{\text{CPU Utilization} = 59.04\%}$$

ⓑ **CPU Utilization if degree of multiprogramming increases to 6.**

Now, $n = 6$

$$U = 1 - (0.8)^6$$
$$= 1 - 0.262144$$
$$= 0.737856$$

$$\boxed{\text{CPU Utilization} = 73.79\%}$$

ⓒ **Interpretation – Which configuration improves system performance and why?**

* When degree of multiprogramming increases from 4 to 6, CPU utilization increases from 59.04% to 73.79%.
* This means CPU stays busy more often and spends less time idle.
* So, higher degree of multiprogramming ($n = 6$) improves system performance.
* More processes in memory help overlap CPU execution with I/O waiting, leading to better utilization.